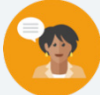


12.3 Writing Algebraic Expressions

Lesson Introduction

Lesson Introduction				
 Benchmark	MA.6.AR.1.1 Given a mathematical or real-world context, translate written descriptions into algebraic expressions and translate algebraic expressions into written descriptions.		 Learning Target	I can translate a written description into an algebraic expression.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.AR.2.1 Translate written real-world and mathematical descriptions into numerical expressions and numerical expressions into written mathematical descriptions.		MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.	
 Essential Questions	- How do you write an algebraic expression from a written description? - How are expressions and equations similar? How are they different? - How can you use models to represent algebraic expressions?			
 Lesson Narrative	The lesson focuses on writing algebraic expressions for mathematical contexts. This is the first experience that students will have with writing expressions including a variable to represent an unknown value. The focus should be on understanding the action of the written description and selecting the appropriate mathematical operation based on that action. The use of bar models will help students make sense of the action and operation.			
 Component Summary	12.3.1 Warm-Up (~5 min.)	Matching and creating diagrams of numerical expressions (<i>SE p. 293</i>)	12.3.4 Collaboration (10 min.)	Matching models and writing expressions (<i>SE p. 297</i>)
	12.3.2 Refresher (5 min.)	Review the four operations (<i>SE p. 294</i>)	12.3.5 Guided Practice (5-10 min.)	Write algebraic expressions from a written description (<i>SE p. 298</i>)
	12.3.3 Guided Instruction (15-20 min.)	Represent algebraic expressions using bar models (<i>SE p. 294</i>)	12.3.6 Wrap-Up (~5 min.)	Reflection (<i>SE p. 298</i>)
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> - Algebraic expression <u>Additional Terms:</u> - Equation	N/A	If you decide to let students use rods or other manipulatives to form bar models, have those ready for students to use.	

 Teacher Tips	Common Misconceptions - Students may want to use keywords to write the expressions and skip the use of the bar models. - Students may struggle with the order of the terms in subtraction expressions. - Students may struggle with the order of the dividend and the divisor in division expressions.	Things to Consider - Although this lesson addresses writing expressions, not equations, it is important for students to think about the action versus keywords. These models will carry over to help students write equations to match word description and also to writing scenarios to match an expression or equation. - Bar models force students to think about the meaning of the words and the actions indicated. This will help students write the algebraic expression rather than strictly translating keywords which often leads to having subtraction or division expressions written in the reverse order. - If time is an issue, consider assigning Guided Practice for homework.							
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard							
	MLR 7: Compare and Connect To foster students’ meta-awareness of different mathematical approaches and representations, teachers should demonstrate thinking out loud as they create and interpret bar models. Emphasize that bar models can be constructed in multiple ways to represent the same expression. Students should be prompted to reflect and respond as they discuss different approaches to creating bar models.	MA.K12.MTR.4.1: Discussion and Reflection Students will be working to create and interpret bar models to represent expressions. When working with a partner, encourage students to clearly communicate their ideas and to collaborate with their partner to analyze each other’s ideas. Students will also justify their responses and recognize and correct errors in collaboration with their partner.							
 Differentiate Instruction	Bar models drive the instruction throughout Lesson 3. Consider providing additional scratch paper and colored pencils or markers for students to engage with more practice as they work on each component in the lesson. Consider having groups construct their own anchor charts on bar models for additional scaffolding.	Equal Part	Equal Part	Equal Part	Equal Part	Equal Part	Equal Part	Equal Part	Equal Part
		Whole							
Lesson Notes:									

12.3.1 Warm-Up: Bell Work

Component Overview: Students engage with visual models as they orient to the content of the lesson. Students begin by calculating the value of given diagrams before producing diagrams of their own for addition and multiplication problems.



12.3.1 Warm-Up: Bell Work

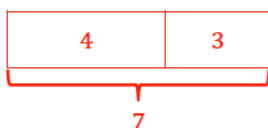
- Two diagrams are shown. One represents $3 + 5 = 8$. The other represents $5 \cdot 3 = 15$.

Complete the table by writing the equation that matches each diagram in the space provided and labeling the length of each diagram.

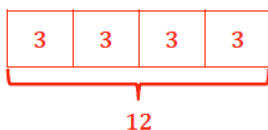
Equation	$5 \cdot 3 = 15$	$3 + 5 = 8$
Diagram		

- Draw a diagram that represents each equation.

a. $4 + 3 = 7$



b. $4 \cdot 3 = 12$



The diagram could also be constructed with three equal segments of 4.



Monitor for and highlight flexibility and fluency in student diagrams. For example, ask one student who drew four groups of three and another who drew three groups of four to display and discuss their equivalence.

12.3.2 Refresher: Sums, Differences, Products, and Quotients

Component Overview: Students review vocabulary related to the four operations. If time allows, discuss how the operations relate to each other.



12.3.2 Refresher: Sums, Differences, Products, and Quotients

1. Complete each statement.

The

☐ difference
☐ product
☐ quotient
☒ sum

 of 4 and 6 is 10 because $4 + 6 = 10$.

The

☒ difference
☐ product
☐ quotient
☐ sum

 of 8 and 3 is 5 because $8 - 3 = 5$.

The

☐ difference
☐ product
☒ quotient
☐ sum

 of 32 and 2 is 16 because $\frac{32}{2} = 16$.

The

☐ difference
☒ product
☐ quotient
☐ sum

 of 7 and 9 is 63 because $7 \times 9 = 63$.



Consider pointing out that the examples on this page are all equations. Discuss the characteristics of an equation. Ask students how equations and expressions differ and how they are similar.

12.3.3 Guided Instruction: Making Sense With Models

Component Overview: Students review the difference in algebraic expressions and equations. They then use bar diagrams to represent expressions. Encourage students to pay particular attention to how a bar model represents each expression. Focus on the action in the descriptions rather than memorizing the types of bar models.



12.3.3 Guided Instruction: Making Sense with Models

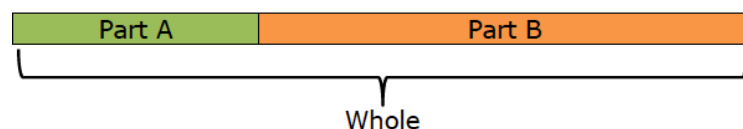
An **algebraic expression** is a mathematical statement containing numbers, operators, and at least one unknown, represented by a variable.

1. Complete the statement.

An expression ☐ does ☒ does not have an equal sign or inequality symbol.

Bar models help connect the action of a problem to an algebraic expression.

Part-Part-Whole Model



The part-part-whole model can be used to show the following actions.

Action	Equation
Combining or joining	$\text{Part A} + \text{Part B} = \text{Whole}$
Taking away or separating	$\text{Whole} - \text{Part A} = \text{Part B}$ $\text{Whole} - \text{Part B} = \text{Part A}$



Position students to connect the visual models of the 12.3.1 Warm-Up to the bar model.



Prompt students to discuss the difference between identifying keywords and focusing on the action of the story. Only identifying keywords can lead to misconceptions; instead, focusing on the action produces more conceptual understanding of real-world problems.

12.3.3 Guided Instruction: Making Sense With Models (continued)

2. "The sum of a number and three" can be shown as the length of the whole bar where one part is n and the other part is 3.

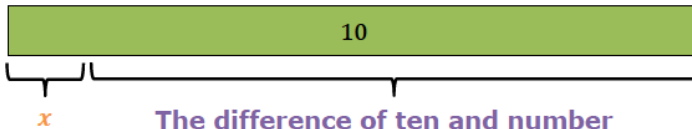
The sum of a number and three



Write an algebraic expression for "The sum of a number and three."

$$n + 3$$

3. "The difference of ten and a number" can be shown where the whole bar is 10 and one part is a number represented by a variable, x . The difference is the other part.



Write an algebraic expression for "The difference of ten and a number."

$$10 - x$$

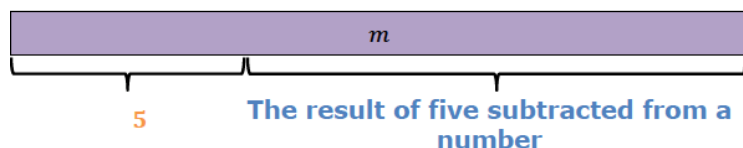


If students are struggling to connect the bar model to an expression, scaffold the instruction further by asking:

- What does "sum" mean?
- What is the whole? What are the parts?
- How can you label the whole on the bar model? How can you label the parts?
- What do you need to do with the parts to find the whole?

12.3.3 Guided Instruction: Making Sense With Models (continued)

4. "Five subtracted from a number" is shown with the whole bar as a number represented by a variable, m , and one part is 5. The other part is the result of 5 being subtracted from the number, m .



Complete the statement.

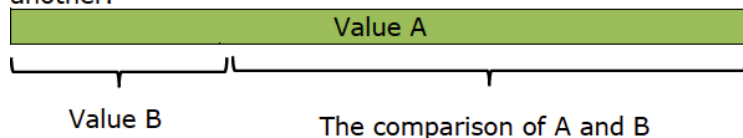
The algebraic expression for "Five subtracted from a number, m " is $m - 5$.



Students may struggle with the lack of detail in the description subtracted from a number. Rephrase to a question "What is five subtracted from your age?" Then, ask students if the answer would be the same if you asked that question to your parents or guardian. Next, discuss where a variable should be used in the expression and why. Ask students how the whole could be represented. Finally, discuss the parts and how they could be represented.

Addition/Subtraction Comparison Model

Comparison models are much like part-part-whole models. They show how much more or less one value is than another.



How much **more** Value A is **than** Value B, or
How much **less** Value B is **than** Value A



Review the meanings of part and whole and use diagrams or images to explain the concepts. Allow students to use different lengths of rods to explore the concept of comparison.

Encourage students to verbalize their comparisons, such as, "The red rod is twice as long as the blue rod."

12.3.3 Guided Instruction: Making Sense With Models (continued)

5. Write an algebraic expression for the bar model shown.



Algebraic expression: $w - 8$.

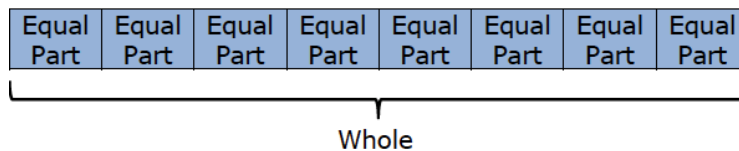


If students are struggling to connect the bar model to an expression, scaffold the instruction further by asking:

- What is the whole?
- What is being taken away?
- In a subtraction problem, where do you write the value being taken away?
- Where do you place the value of the whole?

Equal Parts – Whole Model

The Equal Parts – Whole Model shows the whole sectioned into equal parts.



The equal parts-whole model can be used to show the following actions.

Action	Equation
Repeated addition (multiplication)	$\text{Equal part} + \dots + \text{equal part} = \text{whole}$ $\text{Equal part} \times \text{number of parts} = \text{whole}$
Equal sharing	$\text{Whole} \div \text{number of parts} = \text{equal part}$
Partitioning	$\text{Whole} \div \text{equal part} = \text{number of parts}$



In the table describing the different types of problems that use the equal parts-whole model, make sure that students understand that the ellipsis indicates all of the equal parts between the first and the last. The number of equal parts will not always be the same.



Consider asking students a context that would fit the equal parts-whole model. Examples could include cutting a cake so that everyone has the same amount.

12.3.3 Guided Instruction: Making Sense With Models (continued)

6. "The product of nine and a number" can be shown as 9 equal parts with size of "a number" represented with the variable, a .

a	a	a	a	a	a	a	a	a
-----	-----	-----	-----	-----	-----	-----	-----	-----

The product of nine and a number

Write an algebraic expression for "the product of nine and a number."

$9a$



Review the meaning of "product" with students. Ask students to provide several examples of problems that involve products. Point out that the product of an algebraic expression may be a number and a variable rather than an actual number, as in a multiplication equation.

7. "The quotient of a number and two" could be shown as "a number" represented by the variable, p , sectioned off into two equal parts. The quotient is the size of each equal part.

A number, p

quotient		quotient	
----------	--	----------	--

"The quotient of a number and two" could also be shown as "a number" represented by the variable p , sectioned off into equal parts with size of 2 units. The quotient is the number of equal parts.

A number, p

2	2	2	2	2	2	2	2	2	2	2	2
---	---	---	---	---	---	---	---	---	---	---	---

What is the algebraic expression for "The quotient of a number and 2?"

$p \div 2$ and $\frac{p}{2}$



Write the expression with both the division symbol and the fraction bar. Have students identify the dividend and divisor in each expression. Question 7 shows two interpretations of the quotient. In a mathematical context, it may be interpreted either way. In a real-world scenario, as in future lessons, the interpretation will depend on the context.

12.3.4 Collaboration: Matching Models and Writing Expressions

Component Overview: Students will work together to match bar models to written descriptions of algebraic expressions. Students may complete the activity with a partner, or students may work independently and then compare their answer with another student's.



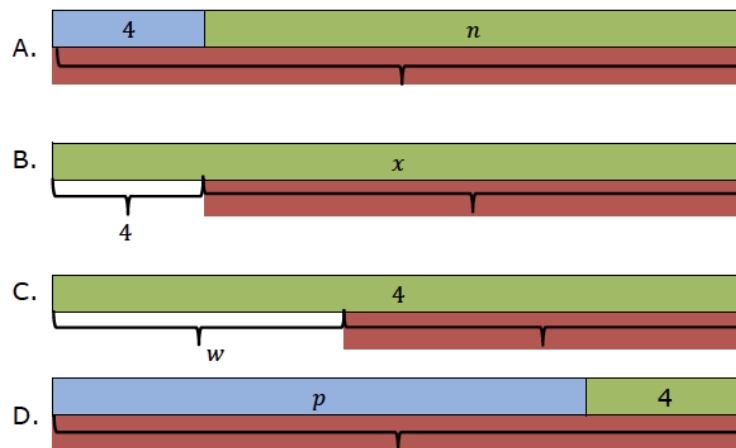
12.3.4 Collaboration: Matching Models and Writing Expressions

Part 1

Work with your partner to match each written description to the appropriate bar model, where the red portion is the result of the written description. Some models will be used more than once.

Complete the table with the letter of the appropriate bar model. Then, write the algebraic expression that matches the word description and bar model.

Written Description	Bar Model	Algebraic Expression
The difference of a number and four	<i>B</i>	$x - 4$
Four decreased by a number	<i>C</i>	$4 - w$
Four more than a number	<i>D</i>	$p + 4$
A number subtracted from four	<i>C</i>	$4 - w$
A number increased by four	<i>D</i>	$p + 4$
Four less than a number	<i>B</i>	$x - 4$
The sum of four and a number	<i>A</i>	$4 + n$



In both parts of the partner work, the same constant is used throughout the problems to make sure students are not just selecting the answer based on that value. Using the same constant value forces students to look at the action and bar model more carefully. Students may interchangeably use A and D for the addition models. In the subtraction models, however, B and C cannot be interchanged.



If students are struggling to match the written descriptions to bar models and algebraic expressions, use guiding questions to further scaffold the activity:

- Why can A and D be used interchangeably? What does this tell us about the commutative property?
- How did you decide whether to use A or D for an addition problem?
- Why can B and C not be used interchangeably? What does this tell us about the operation of subtraction?
- How did you figure out which model represents which subtraction expression?

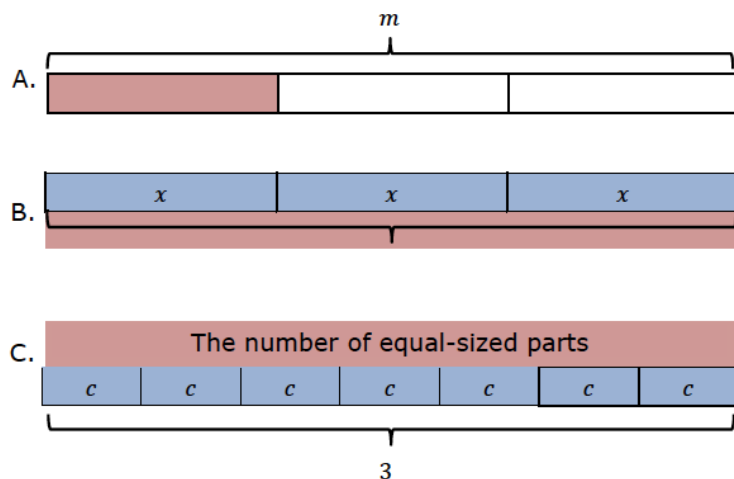
12.3.4 Collaboration: Matching Models and Writing Expressions (continued)

Part 2

Work with a different partner to match each written description to the appropriate bar model, where the red portion is the result of the written description. Some models may be used more than once.

Complete the table with the letter of the appropriate bar model. Then, write the algebraic expression that matches the word description and bar model.

Written Description	Bar Model	Algebraic Expression
A number tripled	<i>B</i>	$3x$
A number equally shared into three parts	<i>A</i>	$\frac{m}{3}$
Three times a number	<i>B</i>	$3x$
The quotient of three and a number	<i>C</i>	$\frac{3}{c}$
The product of three and a number	<i>B</i>	$3x$
A number divided by three	<i>A</i>	$\frac{m}{3}$



Students may struggle with the order of division expressions and/or with writing algebraic expressions. Prompt with questions related to the models, such as:

- “What is the whole?”
- What is the size of the equal parts?
- How many parts?”



Challenge students to create alternative models to represent the algebraic expressions. Students’ models may use bars or other visual tools.

12.3.5 Guided Practice: Writing Algebraic Expressions From Written Descriptions

Component Overview: Students will practice creating bar models and writing algebraic expressions to represent written algebraic descriptions. Students' bar models do not all have to look the same as long as each model accurately represents the action described.



12.3.5 Guided Practice: Writing Algebraic Expressions from Written Descriptions

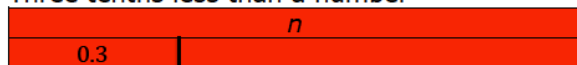
- Sketch a bar model and write an algebraic expression for each written description.

- The product of a number and five



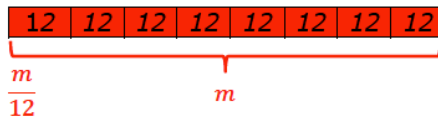
$$5n$$

- Three tenths less than a number



$$n - 0.3$$

- The quotient of a number and twelve

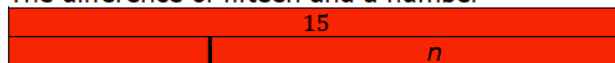


- A number doubled



$$2n$$

- The difference of fifteen and a number



$$15 - n$$

- Compare your expressions and bar models with your partner. Make any revisions if necessary.



Students' bar models may vary. The goal is not to have the exact same models, but to use the models to think about the meaning or action of the written descriptions.



Consider allowing students to work with a partner to draw the models. Student pairs can then compare and discuss their models with those created by another pair.

12.3.6 Wrap-Up: Reflection

Component Overview: Students will assess their understanding of creating an algebraic expression to represent a written description. Encourage students to thoroughly explain what they do and do not understand from the lesson.



12.3.6 Wrap-Up: Reflection

- Choose the emoji that reflects your understanding of translating a written description into an algebraic expression.

Answers will vary.



0 1 2 3 4 5 6 7 8 9 10

- Complete the statement.

I chose this emoji because _____

Answers will vary. Encourage students to respond in detail regarding what they do and do not understand regarding the lesson.



There is no right or wrong answer to the Reflection. Encourage students to thoughtfully assess their understanding of the lesson and to provide a concrete example as part of their explanation of why they chose that emoji.